Dashboard / Kurse / TNF / Institut für Machine Learning / 2019W365105 / Deep Learning and Neural Nets: Exam 12.06.2020 / Exam

Begonnen am	Friday, 12. June 2020, 15:31
Status	Beendet
Beendet am	Friday, 12. June 2020, 16:53
Verbrauchte Zeit	1 Stunde 22 Minuten

Frage **1**
Vollständig
Erreichbare Punkte: 5,00

When are the following regularization techniques applied?

Weight Sharing	During Training
Growing	During Training
Data Augmentation	During Training
Multi-task learning	During Training
Weight Decay	During Training
Dropout	During Training
Pruning	After Training
Self-regularization by SGD	During Training
Neuron noise and weight noise	During Training

Frage **2**
Vollständig
Erreichbare Punkte: 2,00

Mark all activation functions that are strictly monotonically increasing!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Tanh
- [x] b. Linear
- [ ] c. SeLU
- [ ] d. Sigmoid
- [ ] e. ReLU

Frage **3**
Vollständig
Erreichbare Punkte: 2,00

Mark all activation functions that are bounded from above and below!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. SELU
- [x] b. Sigmoid
- [x] c. Tanh
- [ ] d. ReLU
- [ ] e. Linear

Frage 4

Vollständig

Erreichbare Punkte: 2,00

Mark all activation functions that map all inputs to strictly **positive** values!

Wählen Sie eine oder mehrere Antworten:

- [ ] a. Linear
- [x] b. ReLU
- [x] c. Sigmoid
- [ ] d. SELU
- [ ] e. Tanh

Frage 5

Vollständig

Erreichbare Punkte: 2,00

Mark all activation functions whose derivative is not continuous!

Wählen Sie eine oder mehrere Antworten:

- [x] a. ReLU
- [ ] b. Sigmoid
- [ ] c. Tanh
- [x] d. SeLU
- [ ] e. Linear

Frage 6

Vollständig

Erreichbare Punkte: 6,00

Write down the recusion formula for delta-errors of the backpropagation algorithm using the notation of the lecture.

$$\delta^{l-1} = \dots$$

Wählen Sie eine Antwort:

- ( ) a.

$$\delta^{l-1} = \delta^{l}\mathbf{W^{l-1}}diag(f'(s^{l-1}))f(s^{l-1})$$

- ( ) b.

$$\delta^{l-1} = \delta^{l-1}\mathbf{W^{l-1}}diag(f'(s^{l-1}))$$

- (•) c.

$$\delta^{l-1} = \delta^{l}\mathbf{W^{l}}diag(f'(s^{l}))$$

- ( ) d.

$$\delta^{l-1} = \delta^{l}\mathbf{W^{l-1}}diag(f'(s^{l-1}))s'^{l-1}$$

- ( ) e.

$$\delta^{l-1} = \delta^{l}\mathbf{W^{l-1}}diag(f'(s^{l-1}))$$

Frage **7**

Vollständig

Erreichbare Punkte: 4,00

As described in the lecture, for single-layer neural networks with quadratic loss, we have the following objective function $R(b) = \frac{1}{2}\sum_{n=1}^{N}(y^n - (\mathbf{w}^T\mathbf{x}^n + b))^2$, where we made the bias parameter b explicit. Which bias parameter optimizes the objective function assuming fixed $\mathbf{w}$?

Wählen Sie eine Antwort:

- a. $b = \frac{1}{2N}\sum_{n=1}^{N}(y^n - (\mathbf{w}^T\mathbf{x}^n))$
- b. $b = \frac{1}{N}\sum_{n=1}^{N}(x^n - (\mathbf{w}^T\mathbf{y}^n))$
- c. $b = \frac{1}{N}\sum_{n=1}^{N}(y^n - (\mathbf{w}^T\mathbf{x}^n))$
- d. $b = \sum_{n=1}^{N}(y^n - (\mathbf{w}^T\mathbf{x}^n))$
- e. $b = \frac{2}{N}\sum_{n=1}^{N}(y^n - (\mathbf{w}^T\mathbf{x}^n))$

Frage **8**

Vollständig

Erreichbare Punkte: 2,00

What is the interpretation of the expression for the bias that you derived in the question before?

Wählen Sie eine Antwort:

- a. The bias is introduced to avoid saturation of the activation function
- b. The bias is a hyperparameter that needs to be set to maximize performance on a validation set
- c. The bias parameter contains all the weights of the neural network
- d. The bias is the mean difference between the prediction to the target
- e. The bias is introduced to make optimization by SGD more stable

Frage **9**

Vollständig

Erreichbare Punkte: 6,00

Perform the 2D Convolution operation $\ast$ with the following image $\mathbf{x}$ and kernel $\mathbf{w}$:

$$\mathbf{x} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

and

$$\mathbf{w} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

Compute $\mathbf{w} \ast \mathbf{x}$!

Wählen Sie eine Antwort:

- a.

$$\begin{pmatrix} 3 & 4 & 3 \\ 2 & 4 & 3 \\ 2 & 3 & 4 \end{pmatrix}$$

- b.

$$\begin{pmatrix} 4 & 3 & 4 \\ 3 & 4 & 3 \\ 2 & 3 & 4 \end{pmatrix}$$

- c.

$$\begin{pmatrix} 4 & 3 & 4 \\ 2 & 4 & 3 \\ 3 & 3 & 4 \end{pmatrix}$$

- d.

$$\begin{pmatrix} 4 & 3 & 4 \\ 2 & 4 & 3 \\ 2 & 3 & 4 \end{pmatrix}$$

- e.

$$\begin{pmatrix} 3 & 3 & 4 \\ 2 & 4 & 3 \\ 2 & 3 & 4 \end{pmatrix}$$

Frage **10**
Vollständig
Erreichbare Punkte: 4,00

What is Dropout?

Wählen Sie eine Antwort:

- a. Dropout is a initialization technique which drops out neurons of the network with a specified probability
- b. Dropout is a regularization technique which drops out neurons of the network with a specified probability
- c. Dropout is a regularization technique that randomly drops out activation functions in a neural network
- d. Dropout is a regularization technique which drops out neurons of the network in the backward pass with a specified probability
- e. Dropout is a regularization technique which drops out samples of a batch with a specified probability

Frage **11**
Vollständig
Erreichbare Punkte: 2,00

What is the problem with Dropout concerning training-mode and test mode?

Wählen Sie eine Antwort:

- a. During test mode the same neurons are dropped out as during training.
- b. During testing there is a bias in the expectation of the pre-activations compared to the expectation during training with dropout.
- c. There is no problem concerning dropout concerning training and test mode.
- d. During testing the outputs of the backward pass need to be rescaled by the the probability that a unit activation is not dropped.
- e. During testing the outputs of the backward pass need to be rescaled by the inverse dropout probability.

Frage **12**
Vollständig
Erreichbare Punkte: 4,00

An independent test set can be used to estimate the generalization error of a machine learning model, True or False?

Wählen Sie eine Antwort:

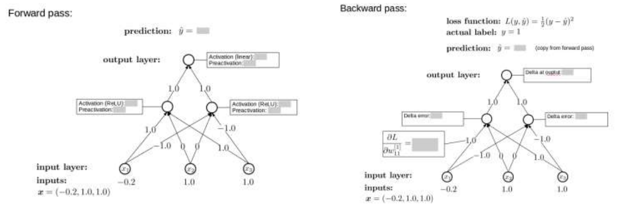
- a. False, the generalization error is estimated directly on the training set
- b. True if the test set is i.i.d it can be used to estimate the generalization error
- c. False, the generalization error cannot be estimated
- d. True, if the model was first trained on the test set
- e. False, the test set needs to be drawn from a different distribution than the training set

Frage **13**

Vollständig

Erreichbare Punkte: 12,00

Compute all outputs, preactivations, activations, and delta errors!



The hidden units of the forward pass are denoted as s_1, and s_2 from left to right. The preactivation of the output layer is denoted as s_3 and the output as $\hat{y}$. The delta at the output is denoted as δ_1, and the hidden deltas as δ_2, and δ_3 from left to right, respectively. Choose the correct value combination!

Wählen Sie eine Antwort:

- a. $s_1 = 0.8, s_2 = -0.8, ReLU(s_1) = 0.8, ReLU(s_2) = 0, s_3 = 0.8,$
- b. $s_1 = 0.8, s_2 = -0.8, ReLU(s_1) = 0.8, ReLU(s_2) = 0, s_3 = 0.8,$
- c. $s_1 = 0.8, s_2 = -0.8, ReLU(s_1) = 0.8, ReLU(s_2) = 0, s_3 = 0.8,$
- d. $s_1 = -0.8, s_2 = 0.8, ReLU(s_1) = 0, ReLU(s_2) = 0.8, s_3 = 0.8,$
- e. $s_1 = 0.8, s_2 = -0.8, ReLU(s_1) = 0.8, ReLU(s_2) = 0, s_3 = 0.8,$

Frage **14**

Vollständig

Erreichbare Punkte: 4,00

For training a machine learning model $g(\mathbf{x}; \mathbf{w})$ with stochastic gradient descent, the important quantity to obtain updates is the gradient of the loss function $\mathcal{L}$ with respect to the inputs $\mathbf{x} : \nabla_x \mathcal{L}(y, g(\mathbf{x}; \mathbf{w}))$, True or False?

Wählen Sie eine Antwort:

- a. True, the gradient w.r.t. the inputs determine the direction in which to go on the loss function to minimize the loss
- b. True, the weights of the model are updated with the gradient of the loss function w.r.t. the input
- c. False, the important quantity is $\nabla_w \mathcal{L}(y, g(\mathbf{x}; \mathbf{w}))$
- d. False, the update is the loss scaled by the learning rate
- e. False, the important quantity is the gradient of the weights w.r.t. the inputs

Frage **15**
Vollständig
Erreichbare Punkte: 4,00

Inputs for a feed-forward neural network should be scaled to the interval [0, 100] - true or false?

Wählen Sie eine Antwort:

- a. True, because gradient descent only works with positive numbers
- b. True, because large numbers leads to larger gradients and therefore faster learning
- c. False, because gradient descent only works with negative numbers
- d. True, because large positive numbers counteract the vanishing gradient problem
- e. False, because because large numbers leads to larger gradients and therefore unstable learning

Frage **16**
Vollständig
Erreichbare Punkte: 4,00

Assume you have a convolutional layer of 3x3 filters with a total of 16 filters and stride 2 that is applied to inputs of dimensions [14,14,6] and you want to initializethe weights of this conv-layer according to \textbf{He's Initialization} scheme. Further assume that you want to use a normal distribution. What is the mean and variance of this normal distribution?

Wählen Sie eine Antwort:

- a. Mean: 1, Variance:

$$\frac{2}{3*3*6}$$

- b. Mean: 0, Variance: $$\frac{2}{3*3*6*16}$$
- c. Mean: 0, Variance: 1
- d. Mean: 0, Variance: $$\frac{2}{3*3*16}$$
- e. Mean: 0, Variance: $$\frac{2}{3*3*6}$$

Frage **17**
Vollständig
Erreichbare Punkte: 4,00

The loss surface of neural networks (loss considered as the function of weights) is in general strictly convex, true or false?

Wählen Sie eine Antwort:

- a. True, because neural networks can approximate any function arbitrarily close according to the universal function approximator theorem.
- b. False, the convexity of the loss function depends on the choice of the optimizer
- c. False, only Sigmoid or Tanh activation leads to strictly convex loss surfaces
- d. False, due to stacking of layers the loss surface increases in complexity
- e. True, due to the stacking of linear layers and activation functions the loss surface is strictly convex

Frage 18
Vollständig
Erreichbare Punkte: 8,00

Suppose we are performing gradient descent on a quadratic objective: $R(\mathbf{w}) = \frac{1}{2} \mathbf{w}^T\mathbf{A}\mathbf{w}$ with symmetric matrix $\mathbf{A}$.The dynamics of gradient descent with learning rate η can be analyzed in terms of the spectral decomposition $\mathbf{A} = \mathbf{Q}\Lambda\mathbf{Q}^T$, where $\mathbf{Q}$ is an orthogonal matrix containing the eigenvalues of $\mathbf{A}$, and $\mathbf{\Lambda} = \mathrm{diag}(\lambda_1, \ldots, \lambda_D)$ is a diagonal matrix containing the eigenvalues of $\mathbf{A}$ in descending order, where $\lambda_1=\lambda_{\mathrm{max}}$ is the largest eigenvalue. Then we have

$\mathbf{w}^t = \mathbf{Q}(\mathbf{I} - \eta\mathbf{\Lambda})^t\mathbf{Q}^T\mathbf{w}^0$.

Based on this formula, what is the value of C such that the gradient descent iterates diverge for $\eta > C$ but converge for $\eta < C$?

Wählen Sie eine Antwort:

- a. $C= \frac{2}{\lambda_{\mathrm{max}}}$
- b. $C= \frac{1}{\lambda_{\mathrm{max}}}$
- c. $C= \lambda_{\mathrm{max}}$
- d. $C=\frac{1}{D}$
- e. $C=1$

Frage 19
Vollständig
Erreichbare Punkte: 3,00

Under which assumption does the optimization procedure for the Perceptron converge?

Wählen Sie eine oder mehrere Antworten:

- a. It converges for all problems as long as the data is normalized
- b. It converges only if the data is in range [0, 1] and linearly separable
- c. It is never guaranteed to converge
- d. It converges for linearly separable problems after finite many steps
- e. It converges for all problems after finite many steps

Frage **20**

Vollständig

Erreichbare Punkte: 2,00

What is the objective function of the perceptron?

Wählen Sie eine Antwort:

- a. \(- \sum_{n=1, y^n=-1}^N y^n \boldsymbol w^T \boldsymbol x^n \)
- b. \(- \sum_{n=1, a^n y^n=1}^N y^n \boldsymbol w^T \boldsymbol x^n \) where \(a^n = \mathrm{sign}(\boldsymbol w^T \boldsymbol x^n) \)
- c. \(- \sum_{n=1, a^n y^n=-1}^N y^n \boldsymbol w^T \boldsymbol x^n \) where \(a^n = \mathrm{sign}(\boldsymbol w^T \boldsymbol x^n) \)
- d. \(- \sum_{n=1, y^n=1}^N y^n \boldsymbol w^T \boldsymbol x^n \)
- e. \(- \sum_{n=1}^N y^n \boldsymbol w^T \boldsymbol x^n \)

Frage **21**

Vollständig

Erreichbare Punkte: 4,00

After optimization of the hyperparameters and architecture on a validation set, the loss on this validation set is an unbiased estimator of the generalization error, True or False?

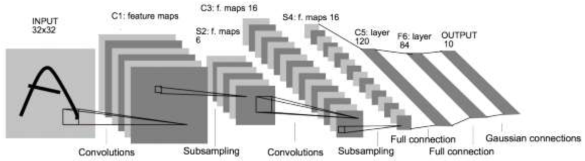
Wählen Sie eine Antwort:

- a. False, since the hyperparameters were tuned on the validation set, which introduces bias
- b. True, hyperparameter tuning on the validation set does not introduce any bias
- c. False, the generalization error is usually estimated by the training set
- d. True, if the validation set is i.i.d. it is an unbiased estimator of the generalization error
- e. False, because the validation set does not fulfill the i.i.d. requirement

Frage **22**

Vollständig

Erreichbare Punkte: 2,00



Layer	Data dimension [width, height, channel] or [#features]	Number of parameters
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters of C1!

The answer should be in format "[width,height,#features],#parameters" or "[#features],#parameters"!

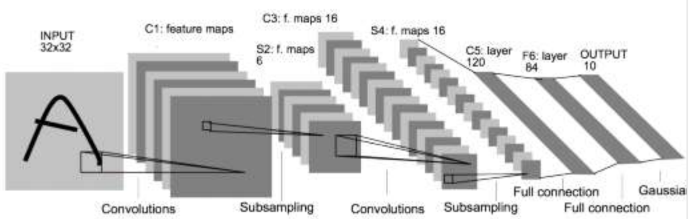
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [28,28,6],156

Frage **23**

Vollständig

Erreichbare Punkte: 2,00



Layer	Data dimension [width, height, channel] or [#features]	Nu par
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters of S2!

The answer should be in format "[width,height,#features],#parameters" or "[#features],#parameters"!

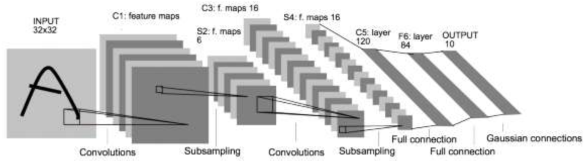
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [14,14,6],0

Frage **24**

Vollständig

Erreichbare Punkte: 2,00



Layer	**Data dimension** [width, height, channel] or [#features]	**Number of parameters**
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters for C3!

The answer should be in format " [width,height,#features],#parameters" or "[#features],#parameters"!

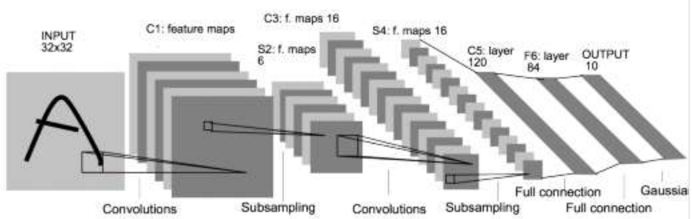
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [10,10,16],2496

Frage **25**

Vollständig

Erreichbare Punkte: 2,00



Layer	**Data dimension** [width, height, channel] or [#features]	**Nu par**
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters for S4!

The answer should be in format " [width,height,#features],#parameters" or "[#features],#parameters"!

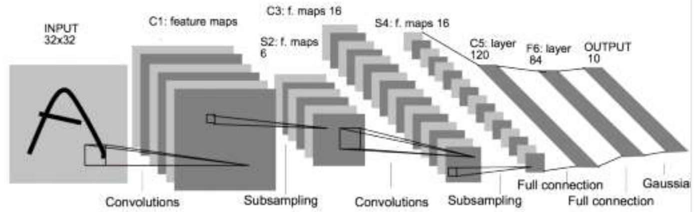
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [5,5,16],0

Frage **26**

Vollständig

Erreichbare Punkte: 2,00



Layer	**Data dimension** [width, height, channel] or [#features]	**Nu par**
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters for C5!

The answer should be in format "[width,height,#features],#parameters" or "[#features],#parameters"!

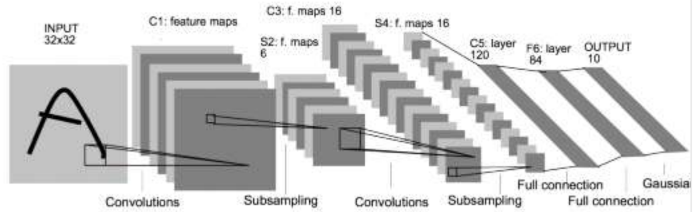
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [1,1,120],49920

Frage **27**

Vollständig

Erreichbare Punkte: 2,00



Layer	**Data dimension** [width, height, channel] or [#features]	**Nu par**
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters for F6!

The answer should be in format "[width,height,#features],#parameters" or "[#features],#parameters"!

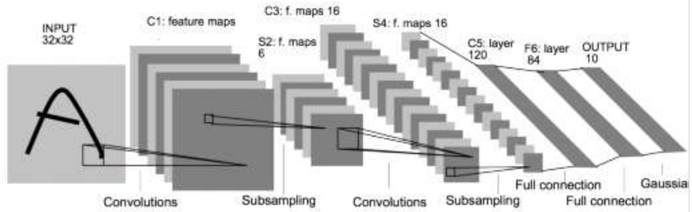
Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [84],10080

Frage **28**

Vollständig

Erreichbare Punkte: 2,00



Layer	**Data dimension** [width, height, channel] or [#features]	**Nu par**
Input layer	[32, 32, 1]	0
C1: 5x5 convolutions with 6 kernels and stride 1		
S2: Max pooling 2x2 with stride 2		
C3: 5x5 convolutions with 16 kernels and stride 1		
S4: Max pooling 2x2 with stride 2		
C5: 5x5 convolutions with 120 kernels and stride 1		
F6: Fully connected layer of 84 units		
O7: Output layer, fully connected with 10 units		

Compute the dimensions and number of parameters for O7!

The answer should be in format "[width,height,#features],#parameters" or "[#features],#parameters"!

Example answer: "[32,32,1],0" (without the "-marks)!

Antwort: [10],840